LAB PYTHON PROGRAMS
CSA1718-ARTIFICIAL INTELLIGENCE

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1. Write the python program to solve 8-puzzle program

PROGRAM:

```
from heapq import heappush, heappop

start = (1, 2, 3,
         4, 0, 6,
         7, 5, 8)

goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

def heuristic(state):
    distance = 0
    for i, tile in enumerate(state):
        if tile != 0:
            j = goal.index(tile)
            distance += abs(i // 3 - j // 3)
            distance += abs(i % 3 - j % 3)
    return distance

def solve(start):
    queue = [(heuristic(start), 0, start, [])]
    visited = set()

    while queue:
        _, cost, state, path = heappop(queue)

        if state == goal:
            for step in path + [state]:
                for i in range(0, 9, 3):
                    print(step[i:i+3])
```

```
            print()
        return

    if state in visited:
        continue
    visited.add(state)

    zero = state.index(0)
    row, col = divmod(zero, 3)

    for dr, dc in [(1, 0), (-1, 0),
                   (0, 1), (0, -1)]:
        r, c = row + dr, col + dc

        if 0 <= r < 3 and 0 <= c < 3:
            j = r * 3 + c
            new = list(state)
            new[zero], new[j] = new[j], new[zero]
            new = tuple(new)

            if new not in visited:
                heappush(queue, (
                    cost + 1 + heuristic(new),
                    cost + 1, new, path + [state]
                ))

solve(start)
```

OUTPUT:

```
(1, 2, 3)
(4, 0, 6)
(7, 5, 8)

(1, 2, 3)
(4, 5, 6)
(7, 0, 8)

(1, 2, 3)
(4, 5, 6)
(7, 8, 0)
```